

Muhammad Yousaf Khalid

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Islamabad, Pakistan (44100)

OBJECTIVE

Results-driven **Computer Science student at NUML, AI/ML Engineer**, and with industrial experience in **Cybersecurity & AI** with a proven track record of deploying scalable predictive models and enterprise-grade web applications. Proficient in **Full-Stack Development (MERN/PERN, .NET Core, Laravel)**, **Machine Learning (TensorFlow, PyTorch)**, and secure **RESTful API** architectures. Leveraging a unique analytical background in **Pre-Medical sciences** and low-level **Assembly logic**, I aim to deliver intelligent, secure, and user-centric solutions at the intersection of AI, Cyber Security, and Data Analytics.

EDUCATION

National University of Modern Languages (NUML)

Bachelor of Science in Computer Science

Expected 2028

Islamabad, Pakistan

- **Relevant Coursework:** Web Engineering, Database Systems, Information Security, Data Structures, Algorithms, Artificial Intelligence
- Developed functional web applications using **.NET MVC and PHP Laravel**, applying MVC architecture to separate concerns and build scalable, maintainable codebases.
- Mastered **Advanced SQL** for complex data manipulation and relational modeling, focusing on database optimization and data integrity—skills actively utilized in **Data Analytics**.
- Bridging the gap between software development and **Cyber Security** by applying Information Security principles to secure web architectures, manage buffer handling, and enforce database access controls.
- Utilizing **Data Science** fundamentals to analyze dataset patterns, leveraging Python and SQL to extract actionable insights from intermediate-level academic and professional projects.

Aziz Bhatti Shaheed Army Public School and College

FSc. Pre-Medical

2021 – 2023

Mardan, Pakistan

- Developed a strong **analytical and scientific foundation**, leveraging rigorous problem-solving skills to transition into computational logic and software development.
- Applied systematic research methodologies and quantitative analysis, which now serve as the basis for work in **Data Science** and **complex SQL database modeling**.

EXPERIENCE

Wise Tech Pakistan

AI/ML Engineer

May 2025 – Present

Rawalpindi, Pakistan

- Designed and deployed scalable **Machine Learning models** using Python, TensorFlow, and PyTorch, achieving a 25% increase in predictive accuracy through rigorous hyperparameter tuning.
- Developed and productionized high-performance **RESTful APIs** using **Flask and FastAPI**, reducing deployment cycles by 40% through automated CI/CD integration.
- Led end-to-end data pipelines, including feature engineering and statistical analysis, while documenting technical workflows to ensure model reproducibility and scalability.

INTERNSHIPS

Safexsolutions

Cybersecurity Intern

Present

Remote

- Assisting in vulnerability assessments and ensuring the implementation of secure software architectures, including Role-Based Access Control (RBAC), JWT authentication, and secure API endpoints.

- Monitoring and analyzing potential threat vectors while gaining hands-on experience with modern network security protocols, system hardening, and defensive strategies.

Quantum Synergy Solutions

Artificial Intelligence Intern

Present

Remote

- Developing and optimizing machine learning models using Python, TensorFlow, and Scikit-learn for predictive analytics and natural language processing tasks.
- Collaborating remotely with cross-functional data teams to build AI-driven features, improving data preprocessing efficiency and model evaluation metrics.

PROJECTS

HireStream (React, Node.js, PostgreSQL) | [Live Link](#)

- Developed a full-stack hiring platform supporting distinct workflows and dynamic dashboards for candidates, recruiters, and interviewers.
- Implemented secure **JWT authentication** and role-based access control middleware to protect backend API routes and frontend views.
- Engineered a seamless applicant tracking pipeline, allowing candidates to monitor application statuses and recruiters to manage job postings, shortlist applicants, and schedule interviews.
- Integrated automated status updates driven by interviewer feedback and ratings, significantly streamlining the final selection process.

Enterprise Web Development (.NET Core MVC)

- Developed a robust, cross-platform web application using **.NET Core MVC**, implementing secure user authentication and dynamic content management.
- Utilized **Entity Framework Core** with a **Code-First approach**, defining complex data models and managing database schema versioning through **EF Migrations**.
- Integrated **SQL Server** for persistent storage, ensuring seamless data synchronization between the application's domain models and the relational database.

Laravel Web Application (PHP)

- Architected a professional web portal using the **Laravel Framework**, leveraging Eloquent ORM and Blade templating for a high-performance user experience.
- Engineered secure API endpoints and optimized back-end logic to handle real-time data management for scalable web operations.

Advanced Database Management System (SQL Server)

- Designed and implemented a complex relational database using **SQL Server**, focusing on normalization, stored procedures, and trigger-based automation.
- Executed performance tuning and query optimization to maintain data integrity and speed for high-volume datasets.

Netflix Recommendation System (EDA & Scikit-learn)

- Engineered a content-based recommender system following extensive Exploratory Data Analysis (EDA) to identify viewing trends.
- Utilized **Scikit-learn** to implement similarity modeling, improving suggestion relevance by 35% on validation test sets.

Fake News Detection (NLP)

- Built a Machine Learning pipeline to classify news articles using **TF-IDF** vectorization and **Logistic Regression**.
- Achieved high precision and recall metrics by preprocessing text data and identifying linguistic patterns indicative of misinformation.

Breast Cancer Diagnosis Model (SVM)

- Developed a predictive classification model using a **Support Vector Machine (SVM)** to identify malignant vs. benign tumors.
- Performed feature scaling and hyperparameter tuning on the Breast Cancer dataset to maximize diagnostic accuracy.

Grocery Store Management System (Tkinter & SQLite)

- Developed a desktop application using **Python (Tkinter)** to provide a complete retail solution for inventory and billing.
- Implemented a real-time stock monitoring system using **SQLite** for local, lightweight, and reliable data persistence.

Low-Level Security & Logic Interpretation (Assembly Language)

- Developed a **Password Authentication System** using Assembly Language, implementing low-level buffer management and comparison logic for secure access control.
- Engineered solutions for **converting high-level language constructs** (C/C++) into Assembly, manually optimizing instruction sets for improved execution efficiency.
- Interpreted and reconstructed **complex program logic** using registers and stack operations, gaining deep insights into memory management and CPU-level instruction cycles.

SKILLS

Backend & Web Development: React, Node.js, Express.js, PostgreSQL, JWT Authentication, .NET Core MVC, Entity Framework (Code-First), PHP (Laravel), C#, SQL Server, MySQL, HTML/CSS, JavaScript, RESTful APIs, MERN/PERN Stack

AI & Data Science: Python, Scikit-learn, TensorFlow, PyTorch, OpenCV, Flask, FastAPI, Streamlit, Pandas, NumPy, Seaborn, Matplotlib, Plotly

Low-Level & Security: Role-Based Access Control (RBAC), Assembly Language (x86), Threat Analysis, Vulnerability Assessment, Buffer Management, Information Security, Cyber Security Fundamentals

Tools, Deployment & Design: Vercel, Vite, Git, GitHub, Jupyter, VS Code, SQL Management Studio, Figma, Adobe Illustrator, Canva

MS Office Suite (Expert): Microsoft Excel (VBA/Macros), Word, PowerPoint, Access, Outlook, OneNote, Teams, Visio

Soft Skills: Team Leadership, Problem-Solving, Communication, Critical Thinking, Public Speaking, Collaboration

REFERENCES

Dr. Fazli Subhan

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